

Sean Luka Girgis

Senior Python / SQL Developer | AWS, Pandas, Large Datasets & Financial Systems

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Professional Summary

Senior Python and SQL technology professional with 20+ years of enterprise experience across Python, Pandas, SQL, AWS data platforms, large dataset processing, data validation, financial-services systems, production troubleshooting, automation, and stakeholder reporting. Strong background building Python-driven data workflows, optimizing SQL/reporting layers, validating large operational datasets, documenting support processes, and working in regulated enterprise environments. Best aligned to Python/SQL roles requiring Pandas/NumPy-style data processing, AWS, advanced SQL, financial-domain data workflows, testing awareness, and production data reliability; Flask/backend application ownership, GitLab/Jenkins/Docker/CI-CD, Java, formal OOP design patterns, and algorithm-heavy software engineering are positioned carefully as adjacent or working-knowledge areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark and SQL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Developed Python workflows for data ingestion, cleansing, enrichment, aggregation, validation, reconciliation, and publishing to reporting consumers in a regulated financial-services environment.
- Used advanced SQL across Oracle, Amazon Redshift, and reporting structures to analyze large datasets, troubleshoot discrepancies, optimize queries, and support business-facing data products.
- Designed AWS-backed data platform components using S3, AWS Glue, Athena, Redshift, EC2/ECS concepts, SQL, and PySpark to support scalable extraction, transformation, querying, and downstream analytics.
- Implemented data quality checks and reconciliation practices across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to improve accuracy, completeness, and consistency.
- Prepared forecasting-ready datasets for Prophet and scikit-learn workflows, supporting capacity planning, trend analysis, and bottleneck prediction across enterprise infrastructure.
- Troubleshoot production data issues by tracing source feeds, Python/Pandas logic, SQL transformations, missing or late data, failed outputs, and downstream reporting discrepancies.
- Documented code behavior, data flows, validation checkpoints, support procedures, and operational runbooks to improve maintainability, team handoff, and repeatable production support.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Investigated large monitoring datasets to identify data quality issues, performance trends, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical performance improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services monitoring environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, profiling, validating, transforming, and distributing performance data, replacing manual reporting workflows with repeatable processes.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and technical documentation used by support, infrastructure, and application teams.
- Provided troubleshooting guidance, training material, onboarding support, and knowledge sharing to improve monitoring and reporting adoption.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration validation for a high-throughput shopping engine moving from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported phased cutover, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, and messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, and troubleshooting for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

Production Data Quality and RCA Workflow

Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, and issue documentation.

Technologies: Python, Pandas, SQL, Data Validation, RCA, Runbooks

HorizonScale — Capacity Forecasting and Reporting

Built Python/Pandas/PySpark workflows to process large infrastructure telemetry datasets, validate outputs, prepare forecasting datasets, and produce actionable reporting for capacity and risk discussions.

Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for ingestion, transformation, curated analytics datasets, query optimization, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, Pydantic schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG